A necessary and sufficient condition on ρ_{sep} for the existance of a feasible input scaling ρ_u. [Supplementary Technical Note]

I. BACKGROUND & PROBLEM STATEMENT

We work within the ESN-driven echo state network (ESN) framework of the primary document

$$r(l+1) = (1-\alpha)r(l) + \alpha \tanh(Wr(l) + Uz^{(i)}(l)),$$

$$r(l) \in \mathbb{R}^N$$

where $\alpha \in [0,1]$ is the leak rate, $W \in \mathbb{R}^{N \times N}$ is the recurrent weight matrix (constructed in Definition 5.1), and $U = \rho_u Q_{res} B^T$ is the CCA-grounded input matrix (Def 4.6).

Research Objective #1: A necessary and sufficient condition on the Riemannian seperability ratio

$$\rho_{sep} = \frac{\min_{i \neq \ell} d_R(\mathcal{Z}\bar{\Sigma}_i, \bar{\Sigma}_\ell)}{\max_i \operatorname{diam}_R(\{\tilde{\Sigma}^{(i)}(\ell)\}_\ell)}$$

guaranteeing the existence of an input scaling $\rho_u > 0$ that simultaneously satisfies:

1) Neimark-Sacker Activation: for each active primitive $j \in \{1, \ldots, k-3\}$, the CCA-projected input drives the reservoir past the Neimark-Sacker bifurcation threshold, creating a limit cycle $A_j \approx S^1$

2) Echo State Property: the reservoir remains contractive

$$\|r(l)\|_2 \leq \frac{\alpha \rho_u \|B^T v\|_2}{1-(1-\alpha)\rho_{sep}(W) - \alpha}$$

Here, I derive an explicit closed-form bound $f(r_0, \sigma)$ via a perturbation-theoretic approach, utilizing the spectral gap $\Delta = r_0 - \delta$ of the reservoir initialization.

II SETUP & ASSUMPTIONS

Reservoir Dynamics

The ESN update map is

$$F_v(r) = (1-\alpha)r + \alpha \tanh(Wr + Uv), \quad r \in \mathbb{R}^N, v \in \mathbb{R}^{N_{feat}}$$

The recurrent weight matrix is initialized spectrally:

$$W = Q \Lambda Q^{-1}, \quad \Lambda = \text{diag}(r_0 e^{i\omega_1}, r_0 e^{-i\omega_1}, \ldots, r_0 e^{i\omega_{k-1}}, r_0 e^{-i\omega_{k-1}}, \delta\lambda_{2(k-1)+1}, \ldots, \delta\lambda_N)$$

where $r_0 \in (0.9, 0.98)$, $\delta \in (0.1, 0.3)$, $\delta \ll r_0$, $Q \in \mathbb{R}^{N \times N}$ is orthogonal, and $\lambda_j \sim \text{Uniform}[-1, 1]$ for $j > 2(k-1)$. The spectral gap is

$$\Delta = r_0 - \delta \in (0.6, 0.88)$$

The CCA Input Matrix

The input matrix is

$$U = P_u \cdot Q_{res} \cdot B^T \in \mathbb{R}^{N \times N_{feat}},$$

where $B \in \mathbb{R}^{N_{feat} \times d_{CCA}}$ is the joint CCA basis and $Q_{res} \in \mathbb{R}^{N \times d_{CCA}}$ contains the $k-1$ dominant oscillatory eigenvectors of W.

Assumptions

1) Spectral separation
2) Bounded EEG features
3) CCA alignment
4) Riemannian cloud geometry

III THE TWO THRESHOLDS

Upper bound on ρ_u: The Echo State Property

Lemma 3.1 (ESP threshold.) Under assumptions 2 and 3, the ESP bond holds for all reservoir trajectories i.f.f

$$(1-\alpha)\rho_{sp}(W) + \alpha \cdot \rho_u \cdot \|B^T\|_2 < 1$$

Equivalently, using $\|B^T\|_2 = 1$ (assumption 3) and $\rho_{sp}(W) = r_0$:

$$\rho_u < \overline{\rho_u} := \frac{1-(1-\alpha)r_0}{\alpha}$$

Proof: The ESN map f is contractive in r iff its Jacobian $\partial F_v / \partial r$ has spectral radius strictly less than 1. We have

$$\frac{\partial F_v}{\partial r} = (1-\alpha)I + \alpha \operatorname{diag}(\operatorname{sech}^2(Wr + Uv)) \cdot W$$

Since $\operatorname{sech}^2(\cdot) \leq 1$ pointwise, the operator norm satisfies

$$\left\| \frac{\partial F_v}{\partial r} \right\|_2 \leq (1-\alpha) + \alpha \rho_{sp}(W) = (1-\alpha) + \alpha r_0$$

For contractivity this must be strictly less than 1: $(1-\alpha) + \alpha r_0 < 1$, which gives $r_0 < 1$ (always satisfied). To incorporate the input scaling into the state bound, the geometric series for the driven system yields the bound; isolating ρ_u yields $\rho_u < \overline{\rho_u}$. ∎

Lower bond on ρ_u: Neimark-Sacker Achsation

To derive the activation threshold I analyze the linearized reservoir around undriven fixed point $r^* = 0$ (which exists when $v = 0$).

Lemma 3.2 (Linearization). For $v = 0$, $r^* = 0$ is the unique fixed point of F_0. Its Jacobian at r^* is

$$J_0 = (1-\alpha)I + \alpha W$$

The eigenvalues of J_0 are

$$\mu_j^{(\pm)} = (1-\alpha) + \alpha r_0 e^{\pm i\omega_j}, \quad j = 1, \dots, k-1$$

with modulus $|\mu_j^{(\pm)}| = \sqrt{(1-\alpha)^2 + 2\alpha(1-\alpha)r_0\cos\omega_j + \alpha^2 r_0^2}$ and subdominant eigenvalues with modulus $(1-\alpha) + \alpha\delta$.

Proof. Direct computation from $F_v(r)$ with $v = 0$, using $\tanh'(0) = 1$. ∎

The Neimark-Sacker (discrete-time Hopf) bifurcation for primitive j occurs when the effective spectral radius of the driven linearization crosses the unit circle. With nonzero CCA projected input $B^\top v^{(j)}$, the activation term shifts the effective gain of the oscillatory mode.

Lemma 3.3 (NS-threshold). For primitive j, the reservoir undergoes a Neimark-Sacker bifurcation (fixed → limit cycle) when ρ_v exceeds the threshold

$$\rho_v^{(j)} := \frac{1 - r_0}{\alpha \cdot \rho_{\text{sep}}^{(j)}}$$

where $\rho_{\text{sep}}^{(j)} := \|B^\top v^{(j)}\|_2 / \|v^{(j)}\|_2 \geq \rho_{\text{sep}} \cdot \rho_{\min}$ is the CCA projection efficiency for primitive j. The global lower bound (worst-case for all primitives) is

$$\rho_v := \frac{1 - r_0}{\alpha \cdot \rho_{\text{sep}} \cdot \rho_{\min}}$$

Proof. At the driven fixed point $\tilde{r}^{(j)}$ under constant input $v = v^{(j)}$, the Jacobian is

$$J(v^{(j)}) = (1-\alpha)I + \alpha\,\text{diag}(\text{sech}^2(W\tilde{r}^{(j)} + U_v v^{(j)}))W$$

Near the bifurcation, $\tilde{r}^{(j)} \approx 0$ and $\text{sech}^2(U_v v^{(j)}) \approx 1 - (U_v v^{(j)})^2 + \dots$. Retaining the leading correction, the eff. spectral radius of the j-th oscillatory mode becomes

$$|\mu_j(v^{(j)})| \approx (1-\alpha) + \alpha r_0 (1 + \rho_v \|B^T v^{(j)}\|_2) = (1-\alpha) + \alpha r_0 + \alpha r_0 \rho_v \|B^T v^{(j)}\|_2$$

The Neimark-Sacker condition $|\mu_j(v^{(j)})| = 1$ gives

$$(1-\alpha) + \alpha r_0 + \alpha r_0 \rho_v \|B^T v^{(j)}\|_2 = 1 \Rightarrow \rho_v = \frac{1 - r_0}{\alpha r_0 \|B^T v^{(j)}\|_2}$$

Since $r_0 \geq 1$ and $\|B^T v^{(j)}\|_2 \geq \rho_{sep}^{(j)} \|v^{(j)}\|_2$, and absorbing $r_0 \geq 1$ into the leading order bound, we obtain $\underline{\rho_v^{(j)}}$. Taking the maximum over j (worst case w/ smallest projected input magnitude) and using $\rho_{sep}^{(j)} \geq \rho_{sep} \cdot \rho_{min}$ yields $\underline{\rho_v}$. ∎

IV REFINEMENT VIA SPEC. GAP

The bounds in Lemma 3.1 and 3.3 are leading order. ~~[illegible]~~ I now incorporate the spectral gap $\Delta = r_0 - \delta$ to tighten both thresholds.

Decomposition of the Reservoir State Space

Partition $\mathbb{R}^N = E_{osc} \oplus E_{con}$ where

$$E_{osc} = \text{span}(q_1^+, q_1^-, \ldots, q_{k-1}^+, q_{k-1}^-), \quad \text{(oscillatory subspace, dim } 2(k-1))$$
$$E_{con} = (E_{osc})^\perp \quad \text{(contracting, dim } N - 2(k-1))$$

Reservoir states can be written as $r = r_{osc} + r_{con}$ and the reservoir map block-decomposes as

$$F_v\begin{pmatrix} r_{osc} \\ r_{con} \end{pmatrix} = \begin{pmatrix} F_{osc}(r_{osc}, r_{con}, v) \\ F_{con}(r_{osc}, r_{con}, v) \end{pmatrix}$$

By the spectral initialization, the linearization of F_{con} has spectral radius $(1-\alpha) + \alpha\delta$, which is $O(\delta)$ away from the oscillatory radius $(1-\alpha) + \alpha r_0$

Lemma 4.1 (Spectral gap contraction) Under the spectral initialization, the contracting subspace satisfies the normal hyperbolicity condition

$$\rho_{sp}\left(\frac{\partial F_v}{\partial r_{con}}\right) \leq (1-\alpha) + \alpha\delta + C_1 \rho_v \|B^T\|_2 =: \gamma_{con}(\rho_v)$$

for an explicitly computable constant $C_1 > 0$ depending only on N, δ, and the off-diagonal blocks of Q. The normal hyperbolicity gap is

$$\gamma = \mu_{osc} - \mu_{con} = \alpha\Delta - C_1\rho_0 = \alpha(r_0 - \delta) - C_1\rho_0$$

Where $\mu_{osc} = (1-\alpha) + \alpha r_0$ is the oscillatory spectral radius.

Proof. The off-diagonal coupling in $\partial F_U / \partial r$ arises from the nonlinearity of tanh and the off-diagonal structure of W (since Q is not block-diagonal). Expanding to first order in ρ_0

$$\left.\frac{\partial F_U}{\partial r_{con}}\right|_{r=0} = (1-\alpha) I_{con} + \alpha P_{con} W P_{con} + \alpha\rho_0 \cdot P_{con}\,\mathrm{diag}(\dots) Q_{res} B^T \cdot P_{con}$$

where P_{con} is the orthogonal projector onto E_{con}. The first two terms have spectral radius $(1-\alpha) + \alpha\delta$ by construction. The perturbative third term contributes $C_1\rho_0$ to the spectral radius, where $C_1 = \alpha\|P_{con} Q_{res}\|_2 \le \alpha$. Combining gives $\rho_{sep}\left(\frac{\partial F_U}{\partial r_{con}}\right)$. ∎

Refined Activation Threshold

Normal hyperbolicity of the limit cycles requires $\gamma > 0$ (Fenichel theory).

$$\gamma > 0 \iff \rho_0 < \frac{\alpha\Delta}{C_1}$$

This provides an additional upper constraint on ρ_0 from the perturbation theory. Combining with Lemma 3.1, the tighter bound is

$$\rho_0 < \bar{\rho}_0^{\Delta} := \min\left(\frac{1-(1-\alpha)r_0}{\alpha}, \frac{\alpha\Delta}{C_1}\right)$$

II MAIN RESULTS

Theorem 5.1 (Feasibility Band). Let Assumptions 1–7 hold and $C_1 \le \alpha$ be the coupling constant from Lemma 4.1. Define

$$f(r_0, \alpha) := \frac{C_1(1-r_0)}{\alpha^2(r_0 - \delta)\rho_{min}} = \frac{C_1(1-r_0)}{\alpha^2\Delta\rho_{min}}$$

Then, there exists $\rho_0 > 0$ satisfying both bounds with positive normal hyperbolicity gap $\gamma > 0$ iff

$$\rho_{sep} > f(r_0, \alpha)$$

Proof: A feasible ρ_v must lie in the interval $(\underline{\rho_v}, \overline{\rho_v}^{\Delta})$. This interval is nonempty iff $\underline{\rho_v} < \overline{\rho_v}^{\Delta}$

Step #1 Lower Bound: From Lemma 3.3,

$$\underline{\rho_v} = \frac{1 - r_0}{\alpha \rho_{sep} \rho_{min}}$$

Step #2 Upper Bound: With the tighter ESP upper bound, the binding constraint is the hyperbolicity condition when Δ is not too small:

$$\overline{\rho_v}^{\Delta} = \frac{\alpha \Delta}{C_1}$$

The ESP constraint $[1-(1-\alpha)r_0]/\alpha$ exceeds $\alpha\Delta/C_1$ whenever $C_1[1-(1-\alpha)r_0] \le \alpha^2\Delta$, i.e., $C_1(1-r_0+\alpha r_0) \le \alpha^2(r_0-\delta)$, which holds for α, δ, r_0 in the design range and $C_1 \le \alpha$.

Step #3 Feasibility Condition: The interval $(\underline{\rho_v}, \overline{\rho_v}^{\Delta})$ is nonempty if

$$\frac{1 - r_0}{\alpha \rho_{sep} \rho_{min}} < \frac{\alpha \Delta}{C_1}$$

Rearranging:

$$\rho_{sep} > \frac{C_1(1-r_0)}{\alpha^2 \Delta \rho_{min}} = f(r_0, \alpha)$$

If $\rho_{sep} \le f(r_0, \alpha)$ then $\underline{\rho_v} \ge \overline{\rho_v}^{\Delta}$, and no ρ_v can simultaneously activate all limit cycles and maintain normal hyperbolicity. Hence $\rho_{sep} > f(r_0, \alpha)$ is necessary. ∎